

KEUN-SOO HEO (허근수)

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Deep Learning, Multi-Modal, Medical AI, Generative Models

EDUCATION & RESEARCH EXPERIENCE

Korea University

Integrated MS-PhD in Department of Artificial Intelligence

• Medical Artificial Intelligence LAB (MAILAB)

• Advisor : Prof. Tae-Eui Kam

Seoul, South Korea

Mar. 2020 –

Jan. 2020 –

Catholic University of Korea

B.S. in Department of Information, Communication and Electronic Engineering

• Image Signal Processing Lab.

• Advisor : Prof. Changwoo Lee

Bucheon, Gyeonggi-do, South Korea

Mar. 2014 – Feb. 2020

Dec. 2017 – Jan. 2020

PUBLICATION (*: Equal contribution)

Keun-Soo Heo, Ji-Wung Han, Soyeon Bak, Minjoo Lim, Bogyong Kang, Sang-Jun Park, Weili Lin, Han Zhang, Dinggang Shen, and Tae-Eui Kam, “Sparsely Labeled fMRI Data Denoising with Meta-Learning-Based Semi-Supervised Domain Adaptation”, *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2025. (Early Accepted, Top 9%)

Bogyong Kang, Sang-Jun Park, Minjoo Lim, Myeongkyun Kang, **Keun-Soo Heo**, Ji-Hye Oh, Hyun Jung Lee, and Tae-Eui Kam†, “Pre-to-Post Operative MRI Generation with Retrieval based Visual In-Context Learning”, *International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2025.

Sang-Jun Park, **Keun-Soo Heo**, Bogyong Kang, Minjoo Lim, and Tae-Eui Kam, "Group-wise Compression and Summarization via LLM-based Ensemble for Chest X-ray Report Generation," *International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2025. (Oral Presentation)

Sang-Jun Park*, **Keun-Soo Heo***, Dong-Hee Shin, Young-Han Son, Ji-Hye Oh, and Tae-Eui Kam, "DART: Disease-aware Image-Text Alignment and Self-correcting Re-alignment for Trustworthy Radiology Report Generation," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

Bogyong Kang, Hyeonyeong Nam, Myeongkyun Kang, **Keun-Soo Heo**, Minjoo Lim, Ji-Hye Oh, and Tae-Eui Kam, "Target-Aware Cross-Modality Unsupervised Domain Adaptation for Vestibular Schwannoma and Cochlea Segmentation", *Scientific Reports*, 2024.

Jun-Mo Kim, **Keun-Soo Heo**, Dong-Hee Shin, Hyeonyeong Nam, Dong-Ok Won, Ji-Hoon Jeong, and Tae-Eui Kam, "A Learnable Continuous Wavelet-based Multi-Branch Attentive Convolutional Neural Network for Spatio-Spectral-Temporal EEG Signal Decoding", *Expert Systems With Applications*, 2024.

Sanghyeon Cho, Bogyong Kang, **Keun-Soo Heo**, Eunjung Jo, and Tae-Eui Kam, "Enhanced Structure Preservation and Multi-View Approach in Unsupervised Domain Adaptation for Optic Disc and Cup Segmentation", *IEEE International Symposium on Biomedical Imaging (ISBI)*, 2024.

Minjoo Lim, **Keun-Soo Heo**, Jun-mo Kim, Bogyong Kang, Weili Lin, Han Zhang, Dinggang Shen, and Tae-Eui Kam, "A Unified Multi-Modality Fusion Framework for Deep Spatio-Temporal-Spectral Feature Learning in Resting-State fMRI Denoising", *IEEE Journal of Biomedical and Health Informatics*, 2024.

Bogyong Kang, Hyeonyeong Nam, Ji-Wung Han, **Keun-Soo Heo**, and Tae-Eui Kam, "Multi-view Cross-Modality MR Image Translation for Vestibular Schwannoma and Cochlea Segmentation", *MICCAI BrainLes Workshop*, 2022.

Keun-Soo Heo, Dong-Hee Shin, Sheng-Che Hung, Weili Lin, Han Zhang, Dinggang Shen, and Tae-Eui Kam, Deep Attentive Spatio-Temporal Feature Learning for Automatic Resting-State fMRI Denoising, *NeuroImage*, 2022.

SCHOLARSHIP & AWARD

KOBRA TRAVEL AWARDS (KOBRA-TA) (₩3,000,000) Jun. 2025

TEACHING EXPERIENCE

Korea University AI training program for LG CNS
Teaching Assistant

• Subject : Data AI	Sep. 2023 – Nov. 2023
• Advisor : Prof. Sejun Park	
• Subject : Machine Learning	May. 2023 – Jun. 2023
• Advisor : Prof. Tae-Eui Kam	